

Chronic Kidney Disease (CKD)

1. Overview and Epidemiology

Chronic Kidney Disease (CKD) is defined as abnormalities of kidney structure or function, present for more than 3 months, with implications for health. CKD affects approximately 10% of the global population — approximately 850 million people. In India, the estimated prevalence is around 800 per million population (pmp).

CKD is largely silent until advanced stages. It is a major risk factor for cardiovascular disease — CKD patients are more likely to die of cardiovascular events than to progress to end-stage renal disease (ESRD). Globally, it is the 12th leading cause of death, and its burden is rising.

In India, diabetes and hypertension together account for over 40% of CKD cases. Other important causes include chronic glomerulonephritis, obstructive uropathy (kidney stones, prostatic hypertrophy), renovascular disease, and 'CKD of unknown aetiology' (CKDu), particularly in farming communities (hot climate, pesticide and agrochemical exposure, dehydration).

2. CKD Classification

GFR (ml/min/1.73m²)	Description
≥90 (with markers of damage)	Normal or high GFR with kidney damage markers
60-89	Mildly decreased
45-59	Mildly to moderately decreased
30-44	Moderately to severely decreased
15-29	Severely decreased
<15	Kidney failure (ESRD) — dialysis or transplant

CKD is also classified by albuminuria (A1: <30 mg/g, A2: 30-300 mg/g microalbuminuria, A3: >300 mg/g macroalbuminuria). The CGA (Cause-GFR-Albuminuria) classification guides prognosis and management.

3. Causes and Risk Factors

Leading causes: Diabetic nephropathy (most common in India and globally) — hyperglycemia damages glomerular capillaries, causing Kimmelstiel-Wilson lesions. Hypertensive nephrosclerosis — chronic high BP causes arteriosclerosis and ischemic nephron loss.

Other causes: IgA nephropathy (most common primary GN worldwide), focal segmental glomerulosclerosis (FSGS), membranous nephropathy, polycystic kidney disease (PKD — genetic, causing bilateral enlarged cystic kidneys), lupus nephritis, obstructive nephropathy.

Risk factors for CKD progression: proteinuria (strongest modifiable predictor), uncontrolled hypertension, poor glycaemic control, smoking, NSAIDs (nephrotoxic — avoid in CKD), contrast media (acute-on-chronic), hyperuricaemia, anaemia, CKD-MBD (metabolic bone disease).

4. Symptoms and Complications

CKD is often asymptomatic until Stage 4-5. Symptoms of uraemia (late-stage): fatigue, pruritus (itching — phosphate and uraemic toxin deposition), anorexia, nausea and vomiting, hiccups, metallic taste, oedema (legs, face), dyspnoea (fluid overload, anaemia, uraemic pleuritis), decreased urine output, confusion (uraemic encephalopathy).

CKD complications: Anaemia of CKD — EPO deficiency + functional iron deficiency. Metabolic bone disease (CKD-MBD): hyperphosphataemia, hypocalcaemia, secondary hyperparathyroidism (high PTH) → renal osteodystrophy, vascular calcification. Metabolic acidosis. Hyperkalaemia (life-threatening cardiac arrhythmia risk). Fluid overload (hypertension, heart failure). Uraemic pericarditis.

5. Diagnosis and Monitoring

Serum creatinine (not reliable in early CKD — muscle mass dependent). eGFR calculated from creatinine (CKD-EPI equation preferred). Cystatin C-based eGFR more accurate especially at GFR 45-60.

Urine: urine albumin-to-creatinine ratio (UACR) — morning spot urine. Urine dipstick for haematuria, leucocytes. Urine protein-creatinine ratio (PCR).

Imaging: renal ultrasound — kidney size (small kidneys = chronic, large = polycystic, infiltrative, or acute-on-chronic), echogenicity, cysts, stones, hydronephrosis.

Blood tests: U&E; (urea, electrolytes, creatinine), bicarbonate, haemoglobin, calcium, phosphate, PTH, uric acid, fasting glucose, HbA1c, lipid profile, hepatitis B/C, HIV screening before transplant.

6. Management of CKD

Treating underlying cause: optimise glycaemic control (HbA1c <7.0%), blood pressure control (target <130/80 mmHg — use ACE inhibitor or ARB as first-line for their nephroprotective effects, especially in proteinuric CKD).

Reduce proteinuria: ACE inhibitors/ARBs are the cornerstone. SGLT2 inhibitors (empagliflozin, dapagliflozin) have proven nephroprotective effect independent of glucose lowering (CREDENCE, DAPA-CKD, EMPA-KIDNEY trials) — now standard of care for proteinuric CKD.

Diet: low-protein diet (0.6-0.8 g/kg/day) in pre-dialysis CKD may slow progression. Low potassium diet in hyperkalaemia. Low phosphate diet + phosphate binders (calcium carbonate, sevelamer) for hyperphosphataemia. Low sodium (<2g/day) for BP control.

Anaemia management: iron stores first (oral or IV iron). Erythropoiesis-stimulating agents (ESAs — erythropoietin, darbepoetin) if Hb <10 g/dL after iron replete. Haemoglobin target 10-11.5 g/dL (higher targets associated with cardiovascular risk).

7. Renal Replacement Therapy

Haemodialysis (HD): blood is filtered through an artificial kidney (dialyser) 3 times per week, 4-5 hours per session, typically at a dialysis centre. Vascular access: arteriovenous fistula (preferred — lower infection rate), AV graft, or central venous catheter (temporary).

Peritoneal dialysis (PD): continuous ambulatory PD (CAPD) or automated PD (APD) using the peritoneum as the dialysis membrane. Home-based, more physiological, preserves residual renal function longer. Preferred in children, young patients, remote areas, diabetes (better volume control).

Kidney transplantation: the optimal treatment for eligible ESRD patients. Best quality of life and survival. Live donor (family member) preferred. Deceased donor (cadaveric) transplant program growing in India. Requires lifelong immunosuppression (tacrolimus + mycophenolate

+ prednisolone).

In India, ESRD care is mainly funded out-of-pocket. Pradhan Mantri National Dialysis Programme provides free dialysis at government hospitals. Organ donation rates in India remain very low (~0.8 pmp vs 20-30 pmp in Spain/USA), limiting deceased donor transplants.

8. Acute Kidney Injury (AKI)

AKI is defined as: rise in serum creatinine ≥ 0.3 mg/dL within 48 hours, or ≥ 1.5 x baseline within 7 days, or urine output < 0.5 ml/kg/hr for ≥ 6 hours.

Causes: Pre-renal (60-70%): hypovolaemia, sepsis, cardiac failure, hepatic failure, NSAIDs/ACEi in low flow states — all reduce renal perfusion. Intrinsic renal: acute tubular necrosis (ATN) — ischaemia, nephrotoxins (gentamicin, contrast, myoglobin), glomerulonephritis, vasculitis. Post-renal (obstructive): stones, prostatic hypertrophy, tumour.

Management: treat underlying cause. Fluid resuscitation for pre-renal AKI. Avoid nephrotoxins. Monitor potassium, bicarbonate, fluid balance. Indications for emergency dialysis (AEIOU): Acidosis severe (pH < 7.1), Electrolyte (K^+ > 6.5 mmol/L refractory), Intoxication (methanol, ethylene glycol, lithium), Overload (refractory pulmonary oedema), Uraemia (encephalopathy, pericarditis, bleeding).

AKI in pregnancy: septic abortion, eclampsia, HELLP syndrome, acute fatty liver of pregnancy. Cortical necrosis unique to obstetric AKI. Obstructive uropathy from gravid uterus. Fetal outcomes depend on gestational age and aetiology.

9. Nephrotic and Nephritic Syndromes

Nephrotic syndrome: heavy proteinuria (> 3.5 g/day), hypoalbuminaemia (< 3 g/dL), oedema, hyperlipidaemia, lipiduria. Causes: minimal change disease (most common in children — responds well to steroids), FSGS (most common adult primary nephrotic syndrome in India), membranous nephropathy, diabetic nephropathy.

Nephritic syndrome: haematuria (red cell casts in urine), proteinuria, hypertension, oliguria, oedema. Causes: post-streptococcal GN (common in children after throat/skin streptococcal infection), IgA nephropathy, lupus nephritis, ANCA vasculitis.

Investigations: 24-hour urine protein or UACR, urine microscopy, serum complement (C3, C4 — low in post-strep GN and lupus), ANA, ANCA, anti-GBM, ASO titre, renal biopsy (definitive diagnosis).

10. Kidney Disease Prevention in India

Primary prevention: control diabetes and hypertension aggressively. Avoid nephrotoxic medications (NSAIDs, certain antibiotics). Adequate hydration in hot weather and physical work. Avoid traditional herbal remedies with heavy metals (lead, mercury, arsenic) — common cause of CKD in India.

Secondary prevention: screen all diabetics and hypertensives annually for CKD with eGFR and UACR. Treat microalbuminuria with ACE inhibitors/ARBs immediately. Early referral to nephrology at GFR <30 or rapid decline.

India-specific: CKDu (CKD of unknown aetiology) in farming communities of Andhra Pradesh, Telangana, Sri Lanka: dehydration, heat stress, agrochemical exposure. Prevention requires occupational measures (adequate hydration, shade, protective equipment) and community screening.

National Organ and Tissue Transplant Organisation (NOTTO) coordinates deceased donor organ sharing. Initiatives to increase deceased donor rates: mandatory discussion of organ donation at ICU admissions, ICD coding for brain death, sensitization of medical professionals and public.